Adaptive Linear Algebra Explainer

Final Project Documentation
CSYE 7375 — Generative AI

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Spring 2026

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1 System Architecture

1.1 Overview

The Adaptive Linear Algebra Explainer is a retrieval-augmented generation (RAG) pipeline that diagnoses a student's conceptual level before generating any explanation. Unlike a standard RAG chatbot, the system routes the student into one of three tiers based on targeted misconception probes, then filters vector search results to match that tier before passing context to a local LLM.

The pipeline has three sequential stages: **Diagnostic** → **Retrieval** → **Generation**.

1.2 Architecture Diagram

1.3 Component Breakdown

Component	Description
Gradio UI	Stateful chat interface. Manages two stages: diagnostic (Q1–Q3) and query. Loads embedding model and ChromaDB once at startup.
Diagnostic Module	Rule-based classifier using keyword matching on 3 free-text responses. Outputs tier (1, 2, or 3) and a misconception string if applicable.
ChromaDB Vector Store	Local persistent store. Holds 159 chunks tagged with `tier`, `source`, and `topic` metadata.
Embedder	`all-MiniLM-L6-v2` via Sentence Transformers. Used both at index time (build_kb.py) and at query time (retrieve.py).
Retrieval	Cosine similarity search with `$and` metadata filter: {tier, topic}. Falls back to tier-only if fewer than 3 results. Returns top-5 chunks.
Prompt Builder	Constructs a system prompt with tier-specific persona description and optional misconception correction instruction.
Mistral 7B (Ollama)	Local inference at `http://localhost:11434/api/chat`. Instructed to use only retrieved context and cite sources.

1.4 Tier System

Tier	Label	Generation Instruction
1	Geometric Intuition	"a beginner who needs plain English and physical intuition — use no formal notation, explain using analogies and real-world examples"
2	Formal Beginner	"an intermediate learner who understands operations but needs conceptual framing — pair every definition with a concrete example"
3	Algebraically Grounded	"an advanced student comfortable with abstract mathematical structures — use precise mathematical language and formal definitions"

2 Implementation Details

2.1 Technology Stack

Layer	Technology
Language	Python 3.12 (venv)
Document parsing	LangChain `PyMuPDFLoader`
Chunking	LangChain `RecursiveCharacterTextSplitter` (chunk_size=400, overlap=75)
Embedding	Sentence Transformers `all-MiniLM-L6-v2` (384-dim vectors)
Vector store	ChromaDB 1.5.8 (local persistent)
Generation	Mistral 7B Instruct via Ollama (HTTP POST, stream=False, timeout=120s)
Interface	Gradio 6.13 (Blocks API, messages format)
Evaluation	RAGAS 0.4.3 with Ollama as judge LLM

2.2 Knowledge Base

The knowledge base is built from four synthetic PDF documents authored in LaTeX, each calibrated to a specific explanation tier:

Source	Tier	Content Focus
`vector_spaces_notes.pdf`	1	Plain-language intuition, analogies, no formal notation. All 15 topics covered without equations.
`intro_linear_algebra_notes.pdf`	2	Formal definitions paired with concrete worked examples. Covers vectors, transformations, eigenvalues, diagonalization.
`systems_row_reduction_notes.pdf`	2	Procedural treatment: Gaussian elimination, rank-nullity, determinants, matrix inverse via row reduction.
`eigen_notes.pdf`	3	Abstract formulation: 8 axioms, rank-nullity theorem, spectral theorem, Jordan normal form, Cayley-Hamilton.

Total after chunking: **159 chunks** across **15 topics**.

Topic	Chunks	Topic	Chunks
eigenvectors	26	row_reduction	8
vectors	33	basis	8
vector_spaces	13	dot_product	8
determinants	15	eigenvalues	9
linear_transformations	12	linear_independence	6
rank	10	orthogonality	5
matrix_inverse	2	matrix_multiplication	3
systems_of_equations	1		

2.3 Diagnostic Module

Three targeted questions probe known misconceptions in linear algebra:

1. *"True or false: for any two matrices A and B, $AB = BA$."*
   Tests commutativity misconception. "True" → Tier 1. "False" with explanation keywords → Tier 3.

2. *"In your own words, what does multiplying a vector by a matrix actually do?"*
   Tests geometric vs. arithmetic framing. Arithmetic keywords → Tier 1; transformation keywords → Tier 2; space/stretch keywords → Tier 3.

3. *"Have you encountered the concept of a vector space before? If yes, how would you describe it?"*
   Tests abstraction level. "No" → Tier 1; procedural description → Tier 2; axiomatic description → Tier 3.

The final tier is the modal score across the three questions (ties go to the lower tier). If Q1 or Q2 triggers Tier 1, a misconception string is appended to the generation prompt.

2.4 Retrieval

The `retrieve_chunks` function queries ChromaDB with:

```
where = {"$and": [{"tier": tier}, {"topic": topic}]}
n_results = 5
```

If fewer than 3 results are returned (e.g., for thin topics like `systems_of_equations`), a fallback query uses only the tier filter. Results are formatted as a numbered list with source and topic attribution, passed directly into the generation prompt.

2.5 Generation Prompt Structure

```
[system]
You are explaining a linear algebra concept to {tier_description}.

IMPORTANT RULES:
1. Use ONLY the information in the retrieved context.
   Do not add any information from outside the retrieved passages.
2. {misconception_instruction}
3. End your explanation by citing which source(s) you drew from.

[user]
Please explain: {user_query}

Retrieved context:
{formatted_chunks}
```

3 Performance Metrics

3.1 Diagnostic Accuracy (Smoke Test)

The end-to-end pipeline was tested against three canonical response sets designed for each tier. All three tests passed:

Test	Diagnostic Responses	Expected	Result
Tier 1	"true"; "multiply rows and columns"; "no"	1	**1** ✓
Tier 2	"false"; "it transforms the vector"; "yes, set of vectors with addition"	2	**2** ✓
Tier 3	"false because transformations dont commute"; "it applies a linear transformation"; "yes, defined by axioms abstractly"	3	**3** ✓

3.2 Retrieval

All three tiers retrieved 5 chunks for the query "What is an eigenvector?" against the `eigenvectors` topic. Tier-to-source mapping was correct:

Tier	Sources Retrieved	Chunks
1	vector_spaces_notes	5
2	intro_linear_algebra_notes, systems_row_reduction_notes	5
3	eigen_notes	5

3.3 Generation Quality (Qualitative)

Three generated explanations for "What is an eigenvector?" were reviewed against the tier calibration rubric:

Tier 1: Used rubber sheet and stretching string analogies. No formal notation. Addressed the commutativity and arithmetic misconceptions identified by the diagnostic. *Assessment: well-calibrated.*

Tier 2: Provided a concrete worked example with a 2×2 matrix, found eigenvalues $\lambda_1 = 3$, $\lambda_2 = 2$, and corresponding eigenvectors. Paired formal notation with step-by-step reasoning. *Assessment: well-calibrated.*

Tier 3: Used abstract formulation ($T : V \to V$, eigenspaces as kernels, Jordan normal form, Cayley-Hamilton). No analogies; formal mathematical language throughout. *Assessment: well-calibrated.*

3.4 Faithfulness (RAGAS)

Quantitative RAGAS faithfulness evaluation requires a running Ollama instance and was designed to run as a batch job via `evaluation/run_eval.py` against 30 queries (10 per tier). The evaluation framework is fully implemented; results will be reported in the final presentation after the batch run completes.

4 Challenges and Solutions

1. **Import path errors from subdirectory scripts.**
   Python adds the script's containing directory to `sys.path`, not the project root. Files in `tests/` and `interface/` could not find sibling packages like `knowledge_base` and `diagnostic`.
   Solution: Added `sys.path.insert(0, project_root)` at the top of each subdirectory script using `os.path.abspath`.

2. **LangChain API breaking change.**
   `from langchain.text_splitter import RecursiveCharacterTextSplitter` was removed in LangChain 1.x. The module moved to a separate package.
   Solution: Updated to `from langchain_text_splitters import RecursiveCharacterTextSplitter`.

3. **Gradio 6.x dropped the `type` parameter on `gr.Chatbot`.**
   The code was written for Gradio 3/4 tuple-based chat history. Gradio 6 uses a dict message format exclusively and removed the `type` argument entirely.
   Solution: Rewrote all history manipulation to use `{"role": "user/assistant", "content": "..."}` dicts throughout `app.py`.

4. **Thin knowledge base (29 chunks from 2-page PDFs).**
   The original synthetic PDFs were too short to provide adequate retrieval coverage. Several topics

(e.g., `systems_of_equations`, `matrix_inverse`) had zero or one chunks, causing retrieval to always fall back to tier-only filtering.
Solution: Rewrote all four LaTeX source documents to 7–8 pages each, covering all 15 topics explicitly. Recompiled PDFs and rebuilt ChromaDB. Total chunks increased from 29 to 159.

5. **Topic keyword priority conflict.**
Chunks containing the phrase "system of equations" were being assigned to `row_reduction` because that keyword appeared earlier in the priority list. The `systems_of_equations` keyword never matched.
Solution: Moved `"system of equation"` above `"row reduction"` in the `_KEYWORD_TOPIC_MAP` list in `chunk.py`.

6. **Python version incompatibility.**
The system default Python was 3.14, which lacked binary wheels for `torch` and several other dependencies.
Solution: Created a virtual environment with Python 3.12 (`/opt/homebrew/opt/python@3.12/bin/python3.1`
`-m venv venv`), which has full wheel coverage for all dependencies.

7. **Disk space constraint.**
Only 5.8 GB of free disk space remained, insufficient to download `mistral:instruct` (≈4.1 GB) alongside existing Ollama models (≈9.6 GB).
Solution: Removed unused Ollama models (`gemma3`, `llama3`, `phi`, `mxbai-embed-large`) to free space before pulling `mistral:instruct`.

5 Future Improvements

1. **Re-explanation logic.**
The current system generates one explanation per query. A student who is still confused has no path forward except to rephrase. A future version would detect unsatisfied follow-up questions, adjust the tier (e.g., step down from Tier 2 to Tier 1), and re-retrieve from a different angle.

2. **Richer Tier 1 sources.**
The Tier 1 knowledge base is the weakest tier. The original plan included 3Blue1Brown's written materials as a geometric-intuition-first source; these were dropped due to licensing constraints. A future version would either obtain proper licensing or author original geometric explainer content (written analogues of the transformation-first approach).

3. **Deeper diagnostic signal.**
Three questions is a shallow probe. Classification errors at this layer corrupt every downstream step. A future version would use a larger validated question bank, apply a lightweight trained classifier instead of keyword rules, and estimate confidence to flag low-signal tier assignments.

4. **Persistent session memory.**
The current system is stateless between sessions. A logged-in student could accumulate a misconception profile over time, allowing the system to track which concepts have been explained, at what tier, and with what result.

5. **Multi-subject extension.**
The architecture is domain-agnostic; the tier system, RAG pipeline, and prompt structure could transfer directly to other STEM gateway courses (calculus, probability, physics) by swapping

the knowledge base and topic list.

6. **Streaming generation output.**
   The current Ollama call uses `stream=False`, so the student waits 30–90 seconds with no feedback. Enabling streaming output would display tokens as they are generated, dramatically improving perceived responsiveness.

7. **Quantitative RAGAS evaluation.**
   The full 30-query faithfulness evaluation is implemented in `evaluation/run_eval.py` but not yet run at scale. Future work would execute this benchmark, analyze per-tier faithfulness scores, and use flagged low-scoring outputs to drive knowledge base improvements.

6 Ethical Considerations

6.1 Equity Motivation

This system was built explicitly as an equity intervention. Private mathematics tutoring costs $60–70 per hour at the college level. For first-generation students and Pell Grant recipients — who research shows have statistically higher DFWI (D, Fail, Withdraw, Incomplete) rates in gateway math courses — sustained tutoring is not a realistic option. The adaptive diagnostic layer this system provides is normally only available through human tutors. By running entirely locally on free, open-source components, the system is accessible to any student with a laptop, at any hour, at zero cost per session.

6.2 Diagnostic Data Privacy

The diagnostic module collects student responses to three targeted questions about conceptual understanding. These responses are used only within the session to determine tier classification. They are not stored, logged, transmitted, or used to train or fine-tune any model. The system requires no account, no name, and no institutional affiliation. Each session is stateless by design.

6.3 Grounding and Hallucination Mitigation

The generation prompt explicitly instructs the model to use only the retrieved context and to cite the source documents it draws from. This grounding constraint limits (though cannot fully prevent) hallucination and makes the output auditable: a student or instructor can trace any claim back to a specific chunk from a specific source document. This is a meaningful reliability improvement over unconstrained LLM responses.

6.4 Source Licensing and Attribution

The synthetic knowledge base documents were authored originally for this project, so there are no licensing obligations. The original proposal identified MIT OCW 18.06 (CC BY-NC-SA), OpenStax (CC BY 4.0), and Axler's *Linear Algebra Done Right* (CC BY-NC 4.0) as candidate sources; these were replaced by synthetic documents to avoid chunking quality issues with externally authored PDFs, not to circumvent attribution requirements. Any future version incorporating external sources would require explicit attribution in every generated response, consistent with the licenses.

6.5 Bias and Limitations

The synthetic knowledge base was authored in English, from a Western academic mathematical tradition, using standard notational conventions. Students whose prior mathematics education used different conventions (e.g., different index notation, applied-first rather than abstract-first pedagogy) may find the explanations less intuitive than intended. This limitation is disclosed here rather than hidden. Additionally, the three-question diagnostic is a shallow signal: a confident-sounding response does not guarantee correct classification, and a misclassified student receives an explanation calibrated to the wrong level. The Diagnostic Validity Rate metric is designed to catch this failure mode, but it remains a real constraint of the MVP.

6.6 Model Transparency

The system uses Mistral 7B Instruct, an open-weight model. Its training data, potential biases, and failure modes are documented by Mistral AI and are not fully auditable by this project. The grounding constraint in the generation prompt mitigates some risks (the model cannot fabricate source material), but instruction-following slippage — where the model drifts from the retrieved context or ignores the tier constraint — remains a known failure mode of local models operating at this scale.

References

[1] Gardner Institute (2018). *Digging Into the Disciplines I: Accounting for Failure.* Koch & Drake.

[2] Every Learner Everywhere (2022). *What Are Gateway Courses and Why Do They Matter to Equity in Higher Ed?*

[3] Every Learner Everywhere (2023). *Concerned About Equity in Higher Ed? Start with the DFWI Rate.*

[4] Reimers, N. & Gurevych, I. (2019). Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. *EMNLP 2019.*

[5] Es, S. et al. (2023). RAGAS: Automated Evaluation of Retrieval Augmented Generation. *arXiv:2309.15217.*

[6] Jiang, A. et al. (2023). Mistral 7B. *arXiv:2310.06825.*